Yash Kishore Maheshwari

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(617) 777-8735 | Boston, MA | Seeking Jan–Aug 2027 Co-op or Summer 2027 Internship

EDUCATION

Northeastern University, Boston, MA — Expected May 2029
B.S. in Computer Science, University Honors Program — GPA: **3.99/4.00** | Dean's List
Coursework: Algorithms, Computer Systems, Systems Security, Databases, Cloud Computing, Object-Oriented Design
Certifications: AWS Certified Solutions Architect – Associate, Google Cybersecurity Professional Certificate

EXPERIENCE

Undergraduate Research Assistant — Jul 2026 – Present
Chain-of-Thought Complexity, Northeastern Khoury Algorithms and Theory Group | *PEAK Award* — *Boston, MA*
- Implemented 10 transformer constructions in Python and NumPy from mathematical specifications, building a verification suite covering **1.7M+** test cases, including exhaustive checks over selected input ranges.
- Audited stated assumptions against what the proofs use, reporting **11** defects, one prompting a new definition.
- Replaced **one-hot positional encoding** with a scheme fitting **2,004** slots into 37 dimensions.
- Backed every reported finding with a re-runnable test in the repo, a standard the faculty lead adopted team-wide.

Teaching Assistant, CS 2100 — Jan 2026 – Present
Northeastern University, Khoury College of Computer Sciences — *Boston, MA*
- Led weekly labs and 5 hours of office hours for **200+** students on data structures, pandas, and graph algorithms, debugging live with them and tying each topic to how it actually gets used in engineering work.
- Evaluated homework and codewalks across 8 assignments with feedback on structure, test coverage, and design.
- Brought recurring misconceptions from office hours to course staff, changing how those topics were re-taught.

Data Visualization Lead — Sep 2025 – Present
Northeastern Electric Racing — *Boston, MA*
- Engineered an unattended post-race pipeline in Python, pandas, and **Matplotlib** that cut processing time **~40%** and renders a frame-synced telemetry overlay (speed, throttle, brake, pack temp) onto race video for driver debriefs.
- Published nercardata to **PyPI** after gathering requirements from 4 engineering subteams, standardizing sensor queries so new members could pull clean telemetry without learning the database schema.
- Built Python plotting tools on the team's three-year-old **MQTT** telemetry stack, making run data more accessible.

PROJECTS

DepWatch: Vulnerability Intelligence Platform | *Python, Airflow, AWS, Gemini API* — Present
- Built the **Airflow** pipeline behind a vulnerability scanner, landing GHSA and OSV data in MinIO and transforming it into a **33,874**-record Postgres corpus matched against a repo's dependency manifest.
- Added a **Gemini API** layer that assesses whether a flagged package is reachable in the calling code, cutting hundreds of version matches to the few worth acting on, then writes each fix as a prompt a coding agent can execute.
- Indexed the corpus for search by description rather than CVE ID, with Qwen3 embeddings in **pgvector** behind an HNSW index. Deployed on AWS ECS Fargate, RDS, and MWAA under Terraform.

ImpactTrail | *Next.js, TypeScript, Prisma, Postgres, Gemini API* — Feb – May 2026
- **Best Overall Winner, HackBeanpot 2026.** Led the **5-person** team and ran the pitch.
- Wrote the prompt layer that turns a resident's complaint into a volunteer quest, feeding the Gemini API every live quest and validating each response against a **Zod** schema built from the Prisma enums.
- Designed a **13-table** Postgres schema and built volunteer-to-quest matching as a **min-cost flow** capped per volunteer and per quest. Set up GitHub Actions CI with type-check, build, and automatic rollback on a failed deploy.

NU Rate-on | *React, Tailwind CSS, Puppeteer, Supabase* — Nov 2025
- Ranks campus dining halls head-to-head on student votes. Retrieved menu JSON from the Cloudflare-protected dining API through **Puppeteer** browser sessions, normalizing halls, meal periods, and per-item nutrients into five tables.
- Wrote the vote and menu APIs on Supabase, with get-or-create ingestion so a daily re-run never duplicates a menu.

SKILLS

Languages & Tools: Python, Java, TypeScript, JavaScript, SQL, React, Next.js, Git, pytest
Backend & Data: PostgreSQL, FastAPI, Node.js, REST APIs, Airflow, pandas, pgvector, RAG, MQTT
Cloud & Security: AWS (ECS, RDS), Docker, Terraform, CI/CD (GitHub Actions), Linux, CVE/CVSS, SBOM